

CSC 317/2504: Computer Graphics

Course web site (includes all course information):

<https://github.com/karansher/computer-graphics-csc317/>

	LEC 0101/2001	LEC 0201
Instructor	Prof. Karan Singh	Joonho Kim
Contact	karan@dgp.toronto.edu	joonho@dgp.toronto.edu
Lecture	Tue 13:00–15:00 W11017	Tue 15:00–17:00 BA1130
Tutorial	Thu 13:00–14:00 MC252	Thu 15:00–16:00 MC254
Office Hours	Tue 11:30–12:30 BA5258	Wed 12:00–13:00 BA2272

TAs

Karran Pandey, Vishnu Nittoor, Yunze Wei, Zhijie Wu, Sophia Yang. csc317tas@cs.toronto.edu

Questions:

on Assignments: Github issues pages

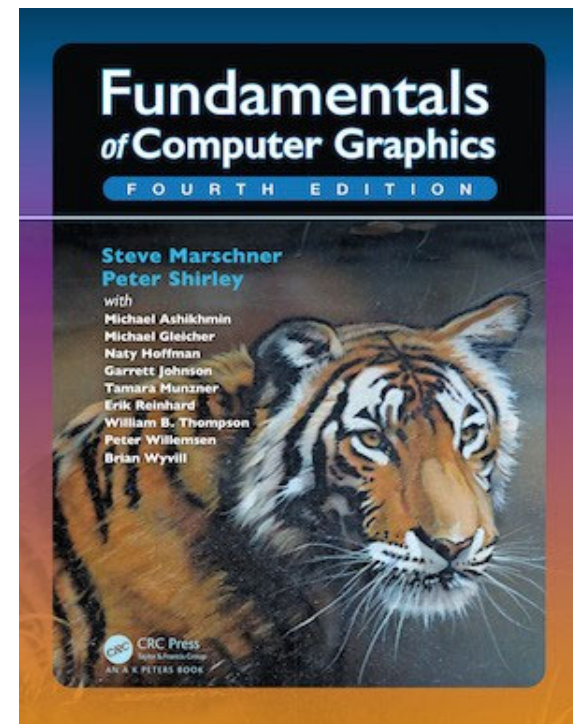
on Administrative stuff: Quercus <https://q.utoronto.ca/courses/395699>

Textbooks: Fundamentals of Computer Graphics

OpenGL Programming Guide & Reference

You will need:

- A computer to install and program the assignments.
- An internet connection for online material and to join any zoom meetings (unlikely).
- Availability during class and tutorial times.
- The textbook for required readings



Schedule (from the webpage)

Week	Sunday	Mon	Tue (Lectures)	Wed (Quiz Release+Due)	Thu (Tutorial+office Hours)	Fri (Assignment Due)
1	Aug 31	Labour Day	Lec 1		Tut 1	
2	Sep 7		Lec 2	q1 released	Tut 2 + oh1	a1
3	Sep 14		Lec 3		Tut 3 + oh2	a2
4	Sep 21		Lec 4	q2 released + q1 due	Tut 4 + oh3	a3
5	Sep 28		Lec 5		Tut 5	
6	Oct 5		refresher+Test 1 (ch1-4)		oh4	a4
7	Oct 12	Thanksgiving	Lec 6	q3 released + q2 due	Tut 6	
8	Oct 19		Lec 7		oh5	a5
9	Oct 26	Reading Week	Reading Week	Reading Week	Reading Week	Reading Week
10	Nov 2		Lec 7+8		Tut 7 + oh6	a6
11	Nov 9		Lec 8 (Drop Day)	q4 released + q3 due	Tut 8 + oh7	a7
12	Nov 16		Guest lecture			
13	Nov 23		Test 2 (ch5-8)+adv. topics	q4 due	oh8	a8+showcase
14	Nov 30		wrap+showcase			

Marking Scheme (from the webpage)

%	Item
64	Assignments (8% each) + 5% extra credit creative showcase
12	4 Quizzes (Online)
12	Test 1 (ch 1-4)
12	Test 2 (ch 5-8)

Academic Honesty Policy is on the webpage and is mandatory reading!

Tutorials sessions will be invaluable for the assignments.

Prerequisites (pre-test)

Background in math, CS and programming is required for this course.

A quick self-monitored pre-test (15-20 mins should be enough to do it)

1. Show you the mathematical background expected and your readiness for this course.
2. Show you what you need to brush up on. Questions about these math operations will not be answered by either Professors or TAs, we expect you to know this stuff.

Discrete Math

Q1: What is the intersection of the interval $\mathcal{A} = [-1, 3]$ and $\mathcal{B} = [0, 4]$? $\mathcal{A} \cap \mathcal{B} =$

Q2: In a balanced binary tree with n leaf nodes, how many internal nodes are there?

Q3: In a balanced binary tree with n leaf nodes, how deep is the tree?
(i.e., number of nodes on a shortest path from root to leaf)

Q4: What is $5!$ (five factorial)?

Q5: How many different ways are there to choose k items from a set of n items?

Linear Algebra

Q6: What size is the result of multiplying a 2×3 matrix and a 3×4 matrix?

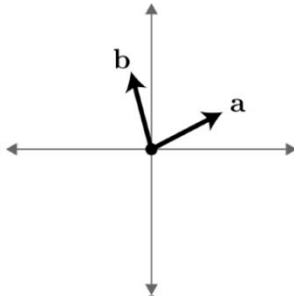
Q7: What is the dot product between the vector $\mathbf{a} = [1, 2, 3]$ and $\mathbf{b} = [4, 5, 6]$? $\mathbf{a} \cdot \mathbf{b} =$

Q8: What is the cross product between the vector $\mathbf{a} = [4, 0, 0]$ and $\mathbf{b} = [0, 0, -2]$? $\mathbf{a} \times \mathbf{b} =$

Q9: What is the matrix product of the matrix $\mathbf{A} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 2 & 1 \\ 0 & -3 & 0 \end{pmatrix}$ and the vector $\mathbf{b} = \begin{pmatrix} -4 \\ 5 \\ 6 \end{pmatrix}$?

$\mathbf{A} \mathbf{b} =$

Q10: Using matrices, express the solution to the following system of equations:
 $3x_1 + 2x_2 - 1x_3 = 10$, $2x_1 - 10x_3 = 0$, $-9x_1 + 9x_2 = -1$. (Form the system, but do not solve).



Q11: Sketch the results of $\mathbf{c} = \mathbf{a} + \mathbf{b}$ and $\mathbf{d} = \mathbf{b} - \mathbf{c}$ on the plot:

Mappings

Q12: Suppose we have a $f : \mathcal{A} \rightarrow \mathcal{B}$, what needs to be true to call f a *well-defined function*?

Q13: Is $f(x) = x^2$ invertible? Why (not)?

Q14: If x and y are real numbers, what sets are the domain and range of $f(x, y) = x^2 + y^2$? (use $\mathbb{R}$)

Logarithms

Q15: Express $\log(a^b)$ in terms of $\log a$ and b . $\log(a^b) =$

Q16: Express $\log(ab)$ in terms of $\log a$ and $\log b$. $\log(ab) =$

Trigonometry

Q17: Express $\cos(a + b)$ in terms of $\cos a$ and $\cos b$. $\cos(a + b) =$

Q18: Given a triangle with side lengths a, b, c and opposite angles A, B, C , express $\sin A$ in terms of a, b , and $\sin B$.
 $\sin A =$

Q19: Given a right triangle with acute angles A and B , express the following in terms of opposite side length a and b :
 $\sin A =$ $\cos A =$ $\tan A =$

Quadratic equation

Q20: How many real-value solutions for x are possible to an equation of the form $ax^2 + bx + c = 0$?
A) zero; B) one; C) two; D) three; E) zero, one or two; or F) zero, one, two or three.

Q21: Identify the solution(s), if any, to $x^2 - 4x - 5 = 0$.

Linear interpolation

Q22: What is *the* linear function $f : [0, 1] \rightarrow \mathbb{R}$, so that $f(0) = a$ and $f(1) = b$?

Q23: What is *the* linear *vector-valued* function $f : [0, 1] \rightarrow \mathbb{R}^3$, so that $f(0) = \mathbf{a}$ and $f(1) = \mathbf{b}$?

Derivatives

Q24: What is the derivative of the function $f(t) = 3t^2$? $\frac{df}{dt} =$

Q25: What is the second derivative of the function $f(t) = 3t^2$? $\frac{d^2f}{dt^2} =$

Q26: What is the partial derivative of the function $f(x, y) = 3x^2 - 3y^2$ with respect to y ? $\frac{df}{dy} =$

Q27: Given a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, how many dimensions does its gradient have?

Q28: Using partial derivative and vector notations express the gradient of a function $f(x, y, z) : \mathbb{R}^3 \rightarrow \mathbb{R}$.
 $\nabla f =$

C++ Programming

Q29: We want to call `func` on `x`, what should be written in place of the question mark in the following C++ code?
A) `*`; B) `&`; or C) nothing.

```
bool func(double & x);  
...  
int main()  
{  
    double * x = new double();  
    func( ? x );  
}
```

Geometry

Q30: What is the volume of a sphere with radius r ?

Today

1. Introduction to Computer Graphics.
2. Preview of Assignments.
3. Raster Images.

Introduction to Computer Graphics

What is Computer Graphics?

Computers:

accept, process, transform and present information.

Computer Graphics:

accept, process, transform and present information
in a visual form.

Ok but... what is the course really about?

The science of turning the rules of geometry, motion and physics into (digital) pictures.

What its not about?

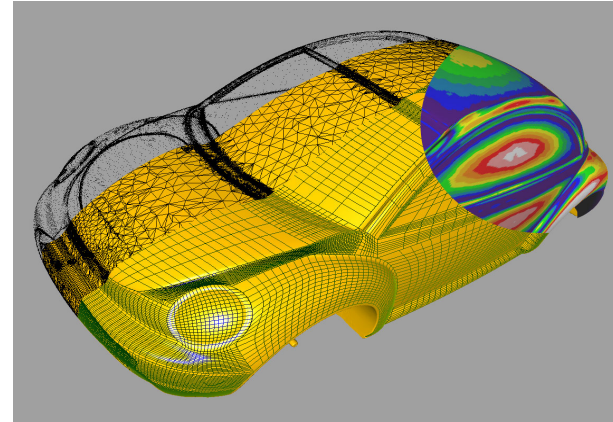
Photoshop, AutoCAD, Maya, Blender, Renderman, Graphics APIs.



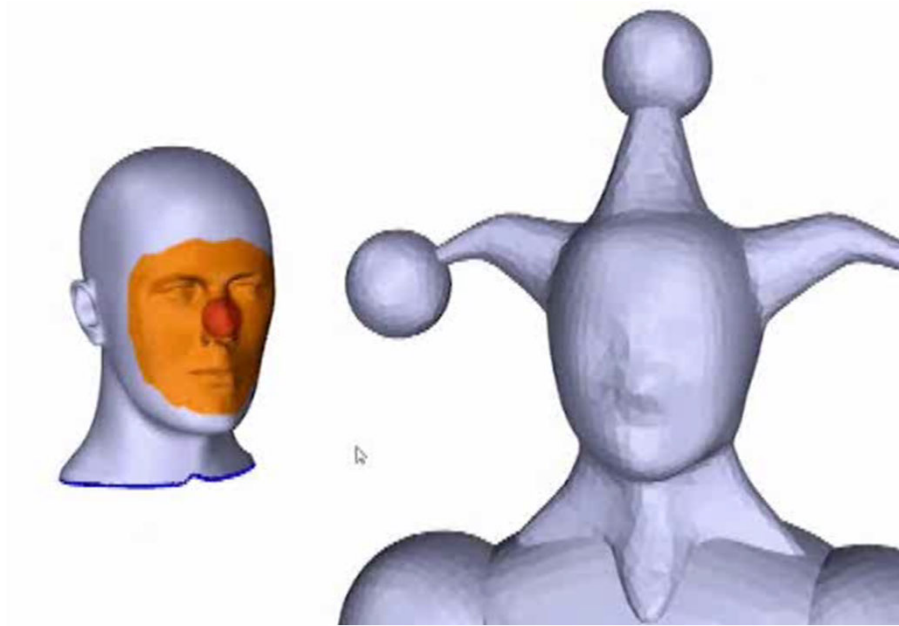
Evan Pan (CSC 317 alumnus) <https://evansamaa.github.io/>

“Core” Areas of Computer Graphics

- Form (modeling)
How do we represent (2D or 3D) objects & scenes?
How do we build these representations?
- Function, Behavior (animation)
How do we represent the way objects move?
How do we define & control their motion?
- Appearance (rendering)
How do we represent the appearance of objects?
How do we simulate the image-forming process?



Modeling

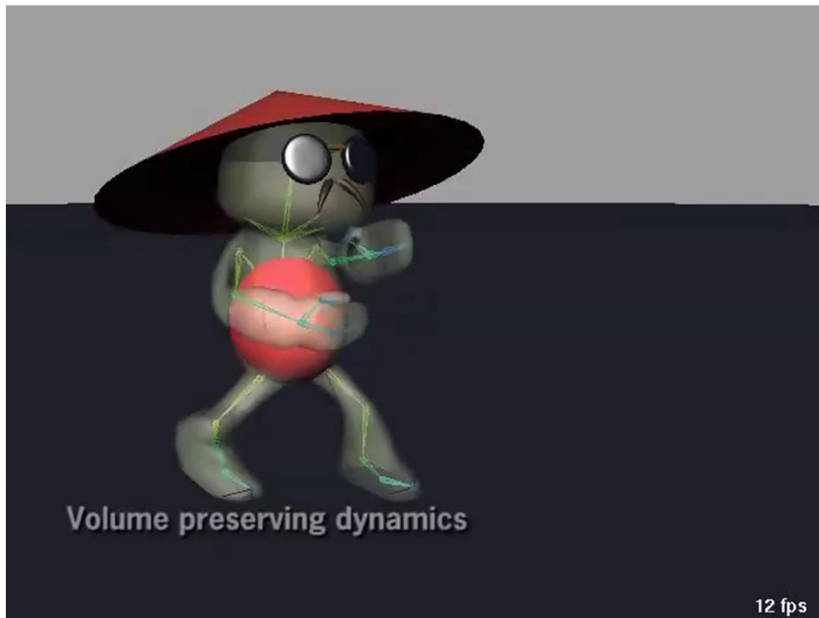


[MeshMixer. Schmidt, Singh, *SIGGRAPH 2010*]
www.meshmixer.com (acquired by Autodesk Inc.)

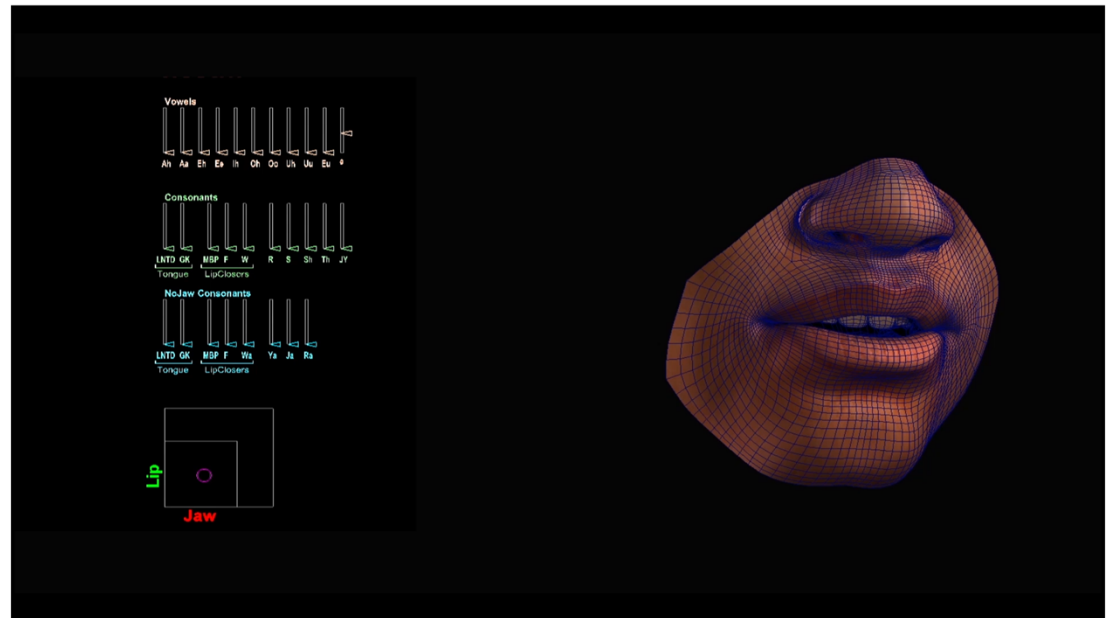


[Face-Extrusion Quadmeshes. Pandey, Baerentzen ,Singh, *SIGGRAPH 2022*]

Animation

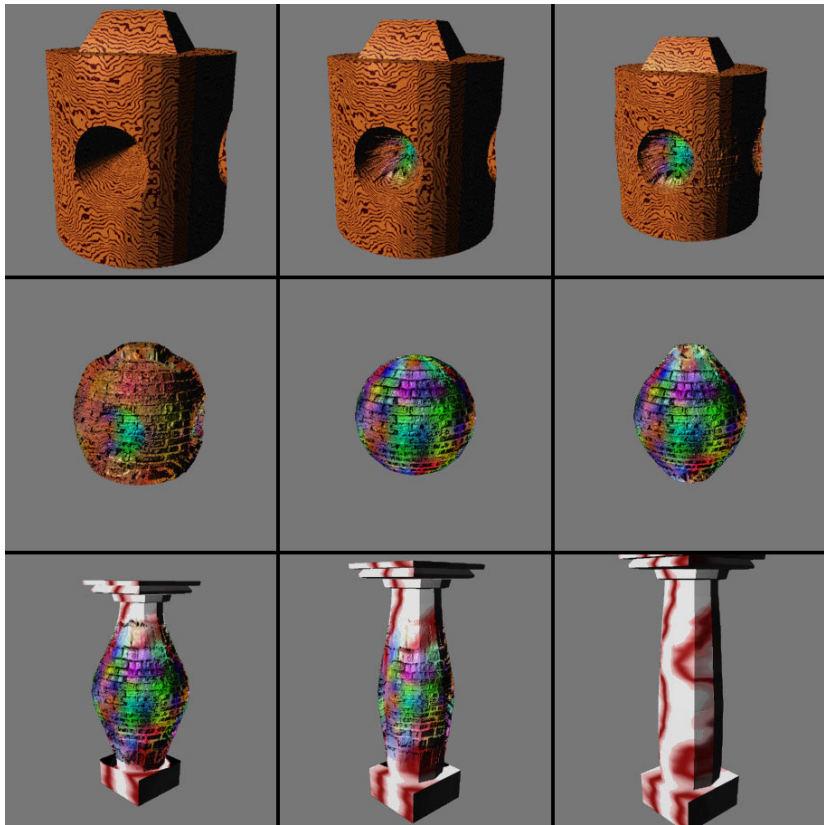


[Kinodynamic skinning using volume-preserving deformations, Angelidis, Singh, *ACM SCA 2007*]



[Visemenet: Zhou et al. *SIGGRAPH 2018*]

Rendering



[Singh, around 1992]



[Weta Digital 2022]

Other Areas of Computer Graphics

User Interaction

Virtual Reality

Visualization

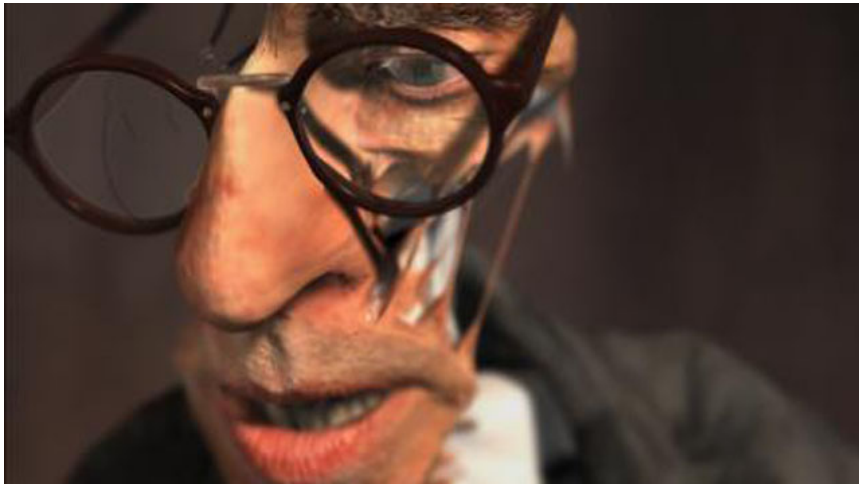
Image Processing

3D Scanning and Fabrication

Computational Photography

Movies

Drive new directions in CG
Set quality standards for CG





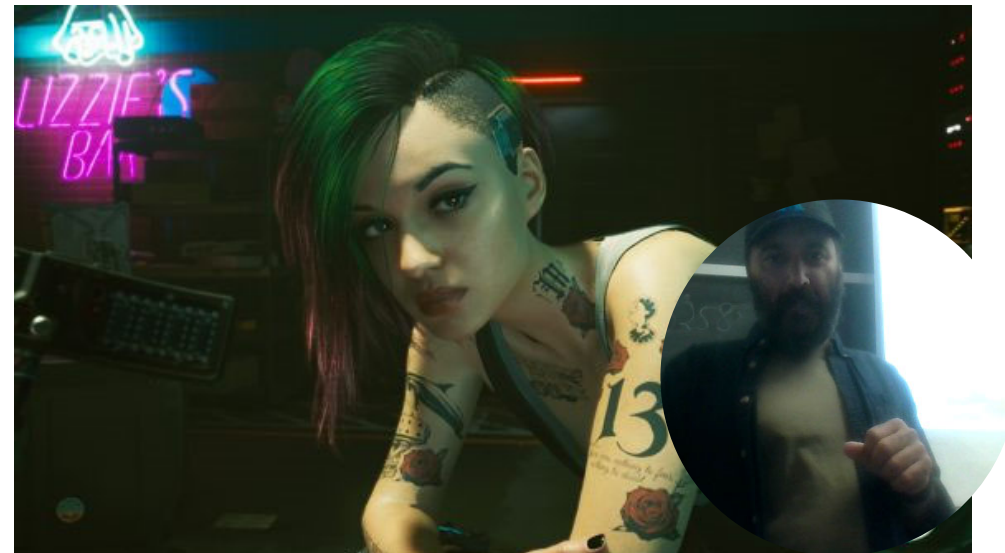
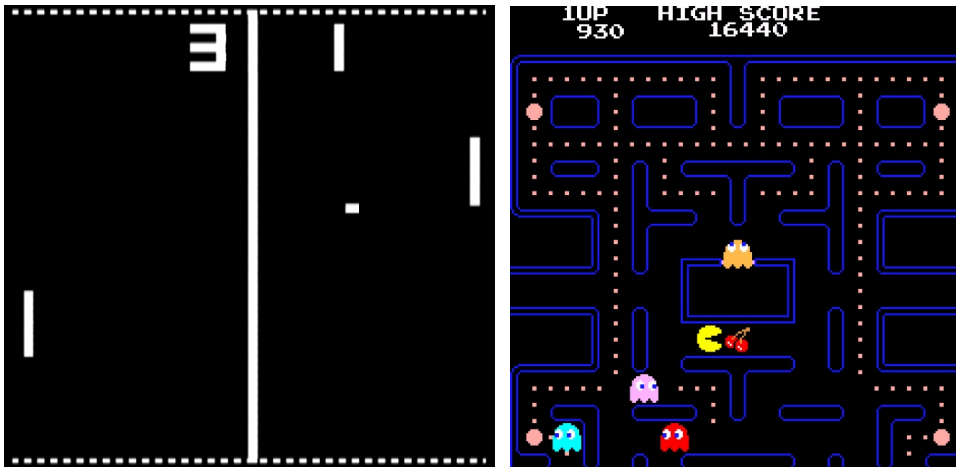
AVATAR

THE WAY OF WATER

Games

Drive interactivity and AI in CG

Push CG hardware to its limits





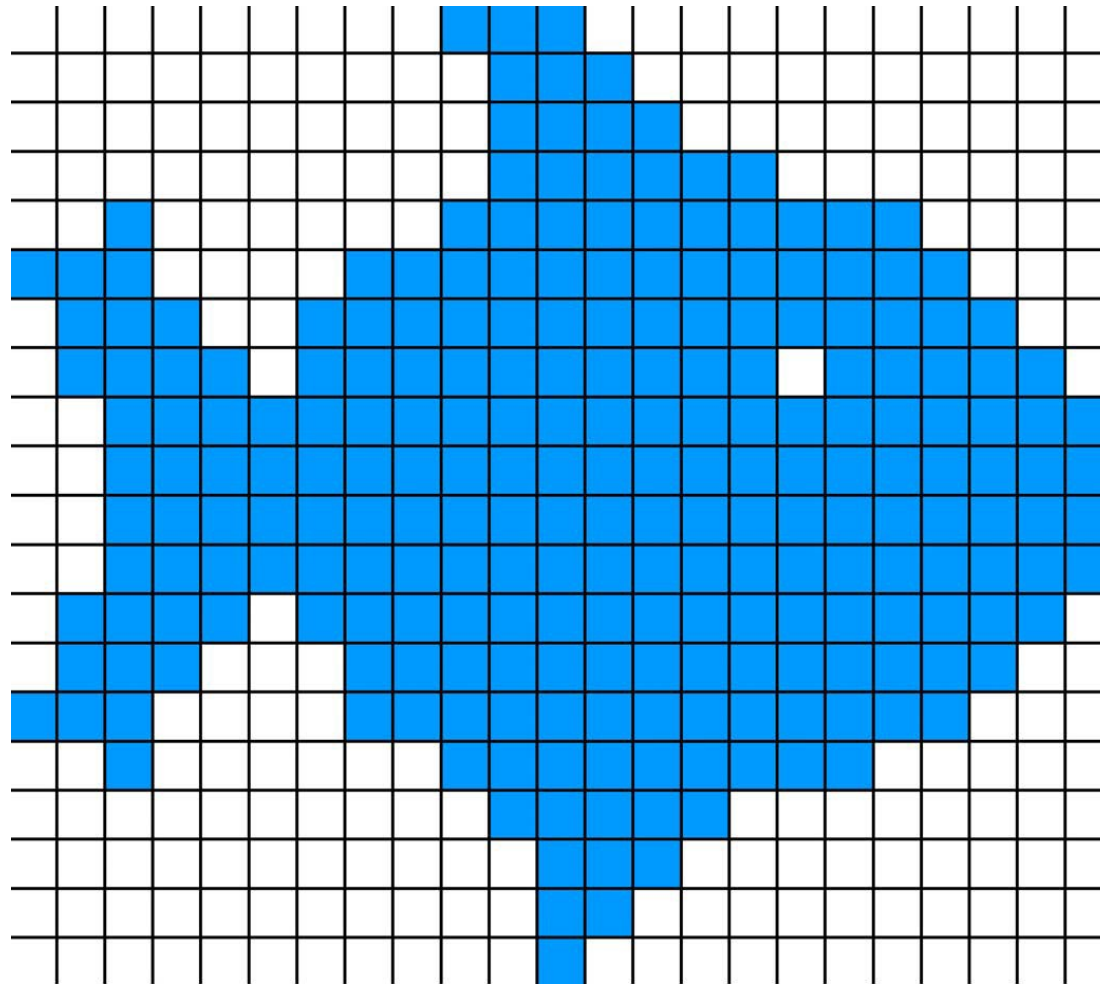
Stellar Blade
Shift Up

Jali

Assignment Previews

1. Raster Images
2. Ray Casting
3. Ray Tracing
4. Boundary Volume Hierarchies
5. Meshes
6. Shaders
7. Kinematics
8. Mass-Springs
9. Generative AI

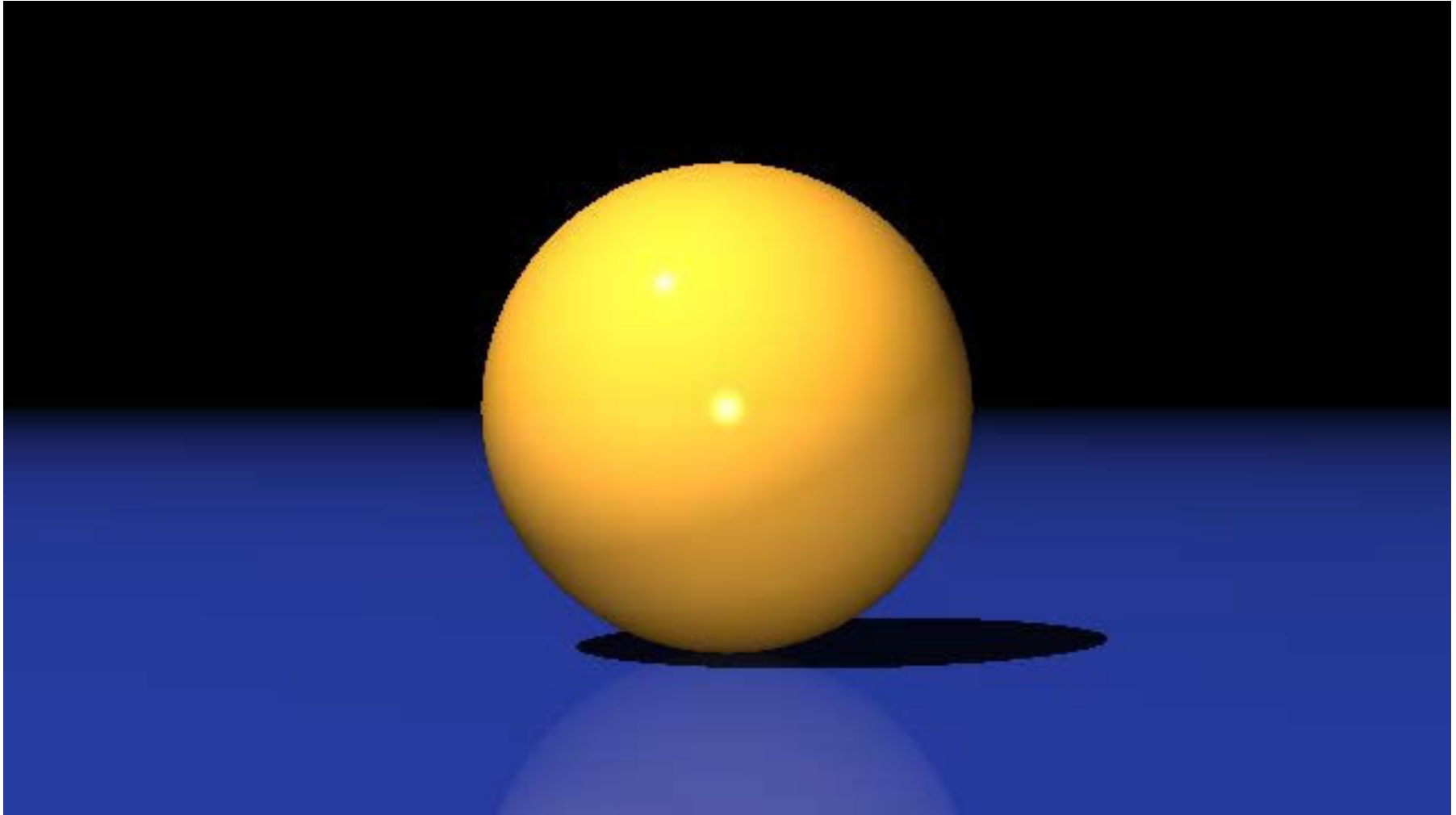
Raster Images



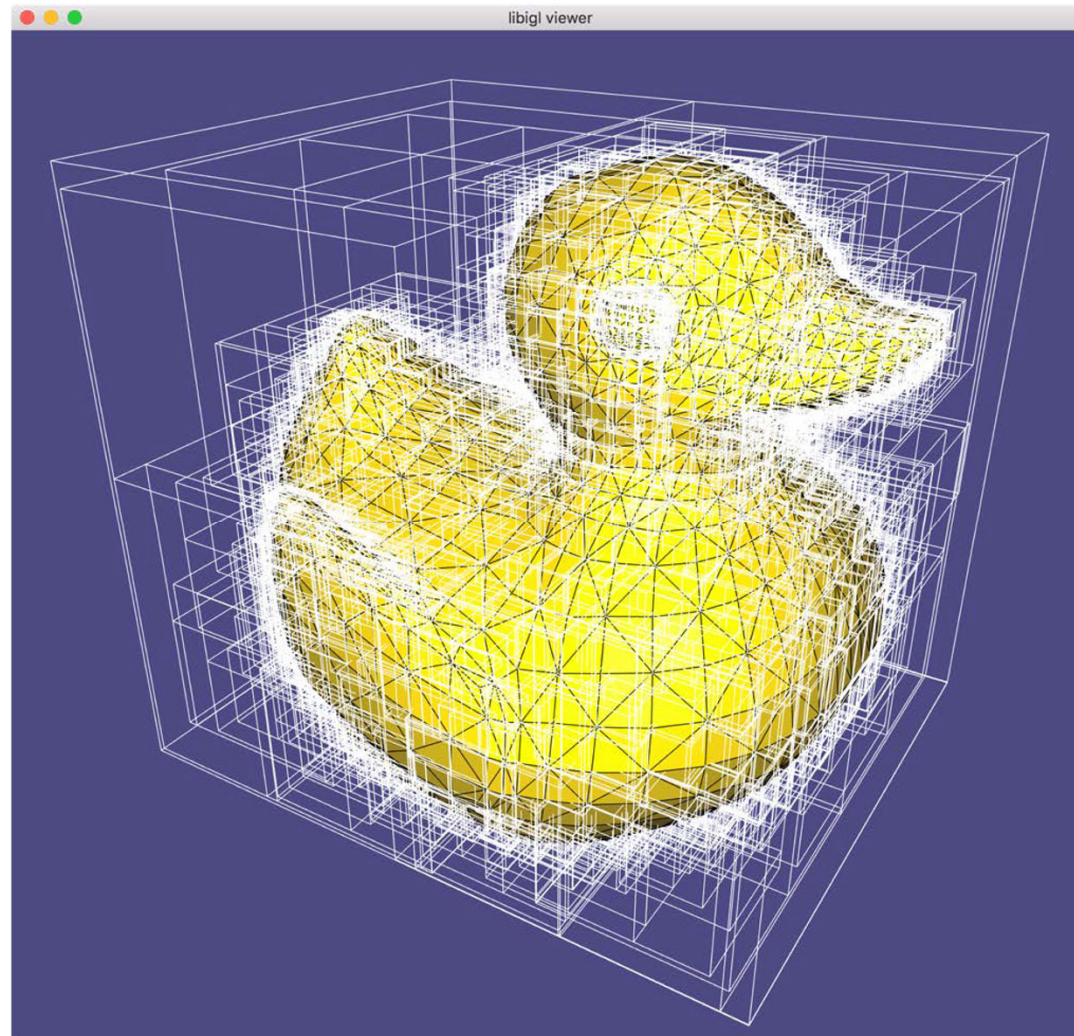
Ray Casting



Ray Tracing

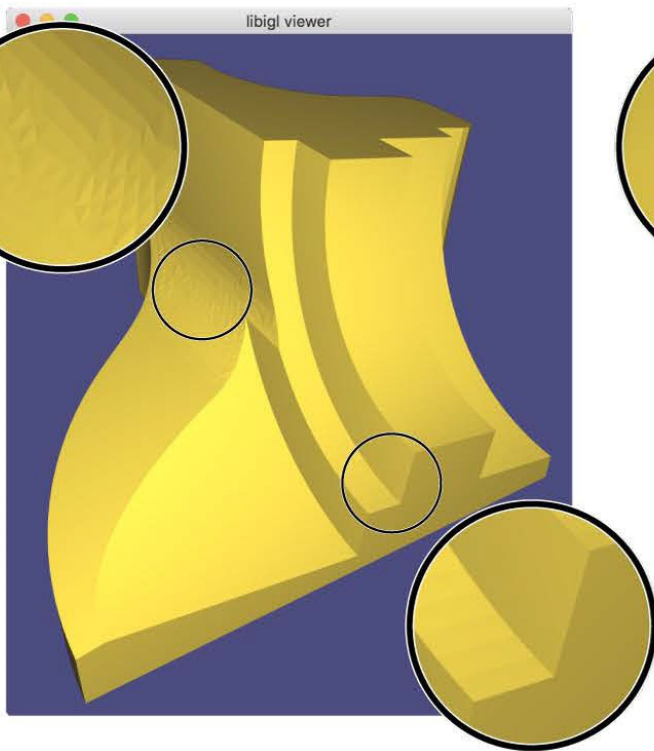


Boundary Volume Hierarchies

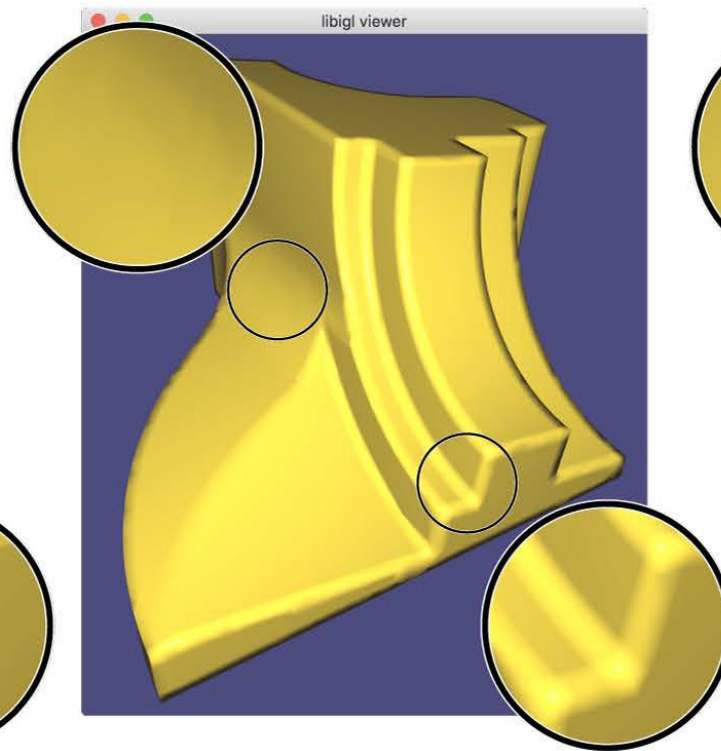


Meshes

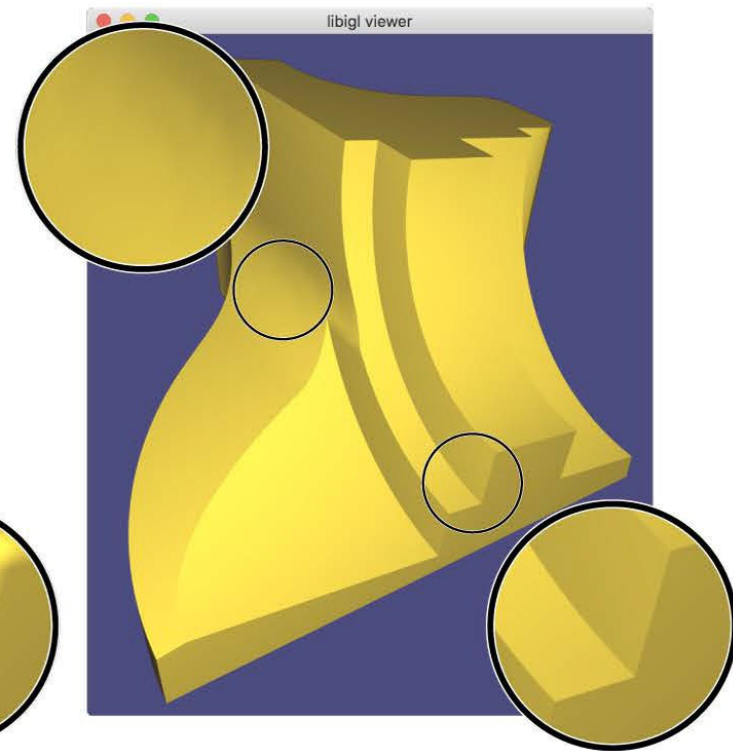
Per-face normals



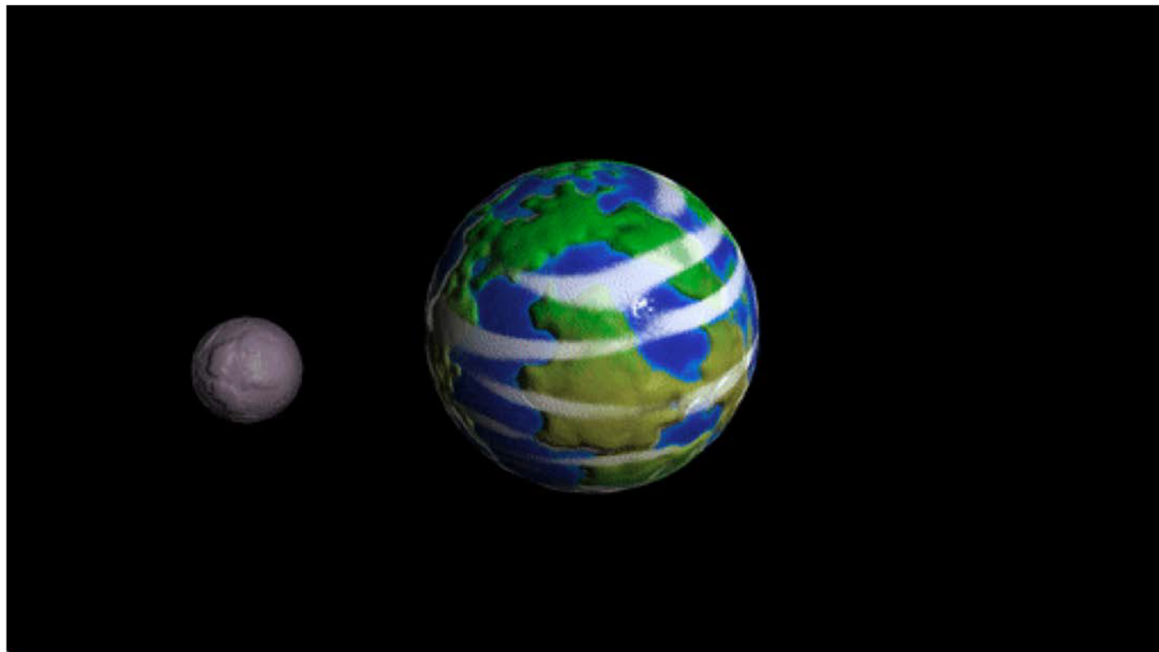
Per-vertex normals



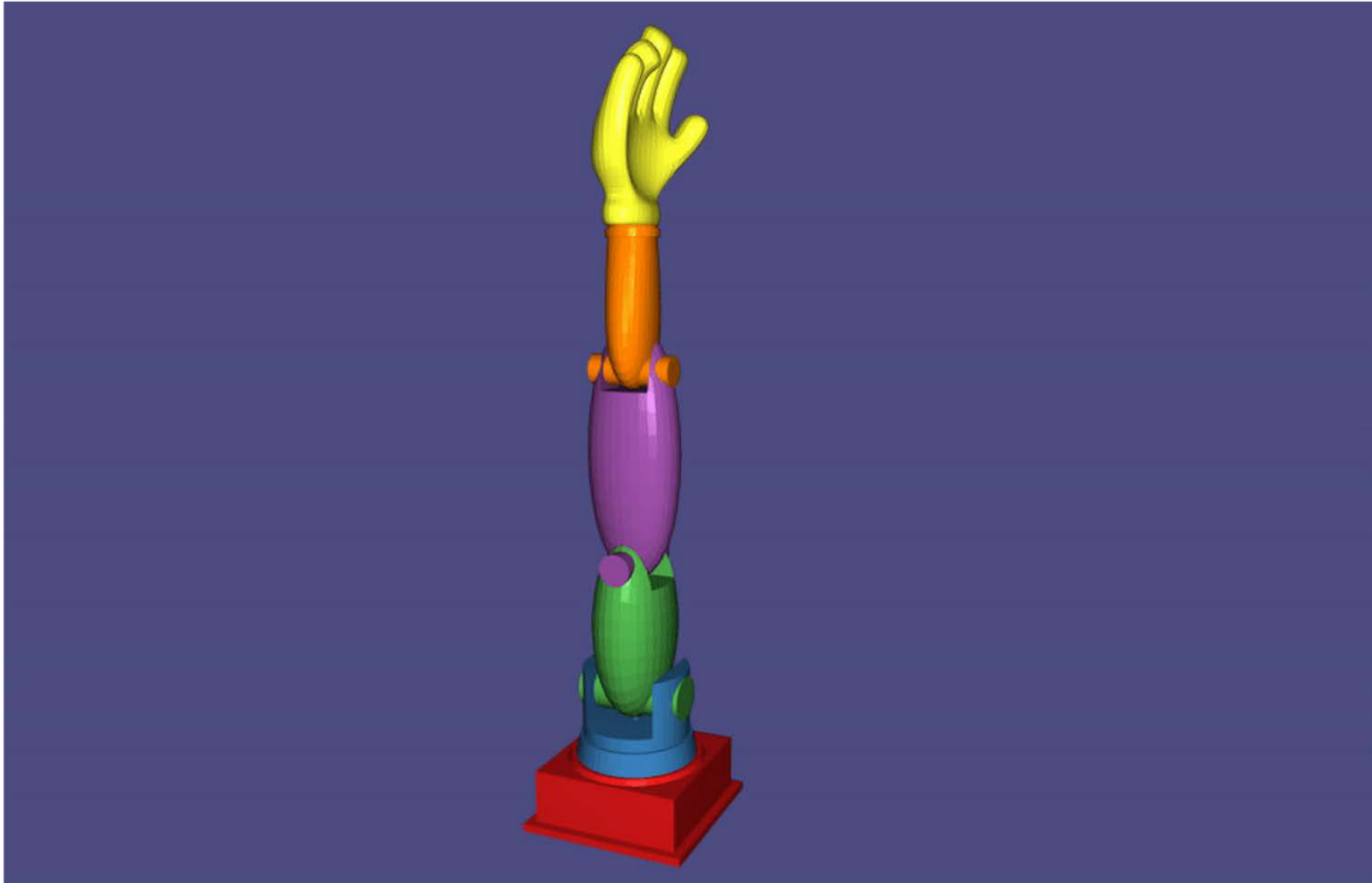
Per-corner normals



Shaders



Kinematics



Mass-Springs



The Wooden Monkey: creative submission for 5% extra credit



**Next lecture >
Raster Images**